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The Generalized Fox–Wright Function: the Laplace Transform, the Erdélyi–Kober Fractional Integral and its Role in Fractional Calculus

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Abstract: In this paper, we consider and study in detail the generalized Fox–Wright function ${}_p\widetilde{\Psi}_q$ introduced in our recent work as an extension of the Fox–Wright function ${}_p\Psi_q$. This special function can be seen as an important case of the so-called I-functions of Rathie and $\overline{H}$-functions of Inayat-Hussain, that in turn extend the Fox H-functions and appear to include some Feynman integrals in statistical physics, in polylogarithms, in Riemann Zeta-type functions and in other important mathematical functions. Depending on the parameters, ${}_p\widetilde{\Psi}_q$ is an entire function or is analytic in an open disc with a final radius. We derive its basic properties, such as its order and type, and its images under the Laplace transform and under classical fractional-order integrals. Particular cases of ${}_p\widetilde{\Psi}_q$ are specified, including the Mittag-Leffler and Le Roy-type functions and their multi-index analogues and many other special functions of Fractional Calculus. The corresponding results are illustrated. Finally, we emphasize the role of these new generalized hypergeometric functions as eigenfunctions of operators of new Fractional Calculus with specific I-functions as singular kernels. This paper can be considered as a natural supplement to our previous surveys "Going Next after 'A Guide to Special Functions in Fractional Calculus': A Discussion Survey", and "A Guide to Special Functions of Fractional Calculus", published recently in this journal.

Keywords: special functions; fractional calculus; Fox–Wright function; generalized Fox–Wright function; Fox H- and Rathie I-functions; Laplace transform; fractional-order integrals; applications

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1. Introduction

The following generalized hypergeometric function was introduced and studied by E.M. Wright in a series of his works from 1933 to 1940 in [1,2], etc., and see also Fox [3], 1928. This is an example of a Fox H-function that in general does not reduce to a Meijer G-function (unless its parameters A_j, B_i are all integers or rational).

Definition 1. *The Wright generalized hypergeometric function ${}_p\Psi_q(z)$, also called the Fox–Wright function, is defined by the power series:*

$$ {}_p\Psi_q\left[\begin{array}{c}(a_1,A_1),...,(a_p,A_p)\\(b_1,B_1),...,(b_q,B_q)\end{array}\middle|z\right]=\sum_{k=0}^{\infty}\frac{\Gamma(A_1k+a_1)...\Gamma(A_pk+a_p)}{\Gamma(B_1k+b_1)...\Gamma(B_qk+b_q)}\,\frac{z^k}{k!}, \tag{1} $$

with arbitrary positive parameters A_j and B_i, complex a_j and b_i, $j=1,...,p, i=1,...,q$, $p\le q$ or $p=q+1$.

For the details and properties of (1), see, e.g., [4–7], etc. Naturally, if $\forall A_j=B_i=1$, this reduces to the more popular ${}_pF_q$-generalized hypergeometric function, which is also

a Meijer *G*-function [8] (Vol.1). It is worth mentioning that a very long list of elementary, classical ([8]) and "fractional" calculus special functions are covered by the Fox–Wright function (see e.q., [9–11], the basic handbooks, among which is [4]).

For readers not so deeply acquainted with these rather general classes of special functions, their numerous examples and their applications, our survey [10] can be recommended as a comprehensive review. Note that, according to the classification proposed by Kiryakova [5] and described again in [10] (Sect. 7, Ths. 7 and 8), almost all the special functions (including the generalized Fox–Wright functions ${}_p\Psi_q(z)$) can be categorized into three main classes: as operators of generalized Fractional Calculus of the three elementary functions, $\cos_{q-p+1}(z)$, $z^\alpha \exp(z)$ and $z^\alpha(1-z)^\beta$, depending on whether $p < q$, $p = q$ or $p = q+1$. This interpretation can be useful for applied scientists and engineers to easily understand the nature and expected behavior of the "complicated" special functions.

The behavior and properties of (1) are characterized by the following basic parameters:

$$\mu = 1 + \sum_{i=1}^{q} B_i - \sum_{j=1}^{p} A_j, \quad R = \prod_{i=1}^{q} B_i^{B_i} / \prod_{j=1}^{p} A_j^{A_j}. \tag{2}$$

The first one indicates when the series in (1) represents an entire function, when it is an analytic in an open disk and when it converges only at the zero point. Namely, series (1) is absolutely convergent in $\mathbb{C}$ if $\mu > 0$; it absolutely converges in the open disk $|z| < R$ when $\mu = 0$, but if $\mu < 0$, the series converges only at zero.

In our recent paper [12], we introduced a generalization of ${}_p\Psi_q$ with arbitrary positive parameters A_j and B_i as in the "classical" case (1), but *with additional "fractional" power parameters* $\alpha_j > 0$ and $\beta_i > 0$ for the Γ-function members in the numerator and denominator, respectively. Then, here, as new contributions, we study its analytical properties, its images under the Laplace transform and FC operators, and its role as an eigenfunction of some new operators of FC.

This ${}_p\widetilde{\Psi}_q$ extension, as well as the other mentioned *I*-functions, are intended to enlarge the family of "Special Functions of Fractional Calculus" (cf. [10]) with Le Roy-type functions, Riemann Zeta-type functions, polylogarithms and other functions important in mathematics and physics.

Definition 2 (Kiryakova, Paneva-Konovska [12])**.** *The generalized Fox–Wright function is defined by the power series*

$${}_p\widetilde{\Psi}_q\left[\begin{matrix}(a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q}\end{matrix}\,\middle|\, z\right] = \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{z^k}{k!}, \tag{3}$$

with arbitrary real or complex parameters a_j, b_i*, positive parameters* A_j*,* B_i *and "fractional" power parameters* $\alpha_j > 0, \beta_i > 0,\ j = 1, ..., p, i = 1, ..., q$.

This new special function is the subject of study in this article. We observe a close analogy between the series representations for the functions (1) and (3), as well as in their behavior. However, while the coefficients in (1) are in general meromorphic functions, those in (3) could be multi-valued due to the fractional powers of the involved Γ-functions. Namely, when some of the parameters α_j or β_i are not integers, the corresponding function $\Gamma^{\alpha_j}(A_j\xi + a_j)^{\alpha_j}$ or $\Gamma^{\beta_i}(B_i\xi + b_i)$ is a multi-valued function of ξ. So, its principal branch should be fixed by drawing a suitable cut, for example, along the negative semi-axis, and supposing that $\Gamma^{\alpha_j}(A_j\xi + a_j)$ or $\Gamma^{\beta_i}(B_i\xi + b_i)$ is positive for all the positive values of ξ.

Let us note that we introduced this generalized Fox–Wright function ${}_p\widetilde{\Psi}_q$ as an important case of the so-called *I*-functions of Rathie [13] and $\overline{H}$-functions of Inayat-Hussain [14], which in turn extend the Fox *H*-functions [3] (see, e.g., [4], etc.); in this scheme, the "classical" Fox–Wright function ${}_p\Psi_q$ has many special functions of Fractional Calculus (cf. [10]).

However, these generalized hypergeometric functions I and $\overline{H}$ also include some Feynman integrals from statistical physics and a number of other important mathematical functions such as polylogarithms and the Riemann Zeta function and its extensions, as well as Le Roy-type functions.

Below, we skip the subtle details on the definitions, Mellin–Barnes-type contours and conditions for the existence and behavior of these rather general and not popular enough special functions, referring to the pioneering works by Rathie [13] and Inayat-Hussain [14], and to the comments in our previous survey [12]. We only recall the formal definitions.

Definition 3 (Rathie [13]). *The I-function is defined as a kind of Mellin–Barnes-type contour integral in the complex plane:*

$$I_{p,q}^{m,n}\left[z\left|\begin{array}{c}(a_j,A_j,\alpha_j)_1^p\\(b_i,B_i,\beta_i)_1^q\end{array}\right.\right]=\frac{1}{2\pi i}\int\limits_{\mathcal{L}}\mathcal{I}_{p,q}^{m,n}(s)\,z^s ds,\quad z\neq 0, \tag{4}$$

where $\mathcal{I}_{p,q}^{m,n}(s)$ *stands for*

$$\mathcal{I}_{p,q}^{m,n}(s)=\frac{\prod\limits_{i=1}^{m}\Gamma^{\beta_i}(b_i-B_i s)\prod\limits_{j=1}^{n}\Gamma^{\alpha_j}(1-a_j+A_j s)}{\prod\limits_{i=m+1}^{q}\Gamma^{\beta_i}(1-b_i+B_i s)\prod\limits_{j=n+1}^{p}\Gamma^{\alpha_j}(a_j-A_j s)}, \tag{5}$$

with the power exponents for the Gamma functions of $\alpha_j>0,\ j=1,...,p$ *and* $\beta_i>0, i=1,...,q$, *that, in general, are not necessarily positive integers. There are three kinds of possible contours* $\mathcal{L}$, *as discussed in [13], similar to those in the definition of the H-function (see [4,15], etc.).*

Another generalization of the H-function, denoted by the symbol $\overline{H}$ and called the *Inayat-Hussain function* ([14]) or *bar-H function*, appears as a slightly simpler case of the I-function:

$$\overline{H}_{p,q}^{m,n}\left[z\left|\begin{array}{c}(a_j,A_j,\alpha_j)_1^n,(a_j,A_j,1)_{n+1}^p\\(b_i,B_i,1)_1^m,(b_i,B_i,\beta_i)_{m+1}^q\end{array}\right.\right]$$

$$=\frac{1}{2\pi i}\int\limits_{-i\infty}^{i\infty}\frac{\prod\limits_{i=1}^{m}\Gamma(b_i-B_i s)\prod\limits_{j=1}^{n}\Gamma^{\alpha_j}(1-a_j+A_j s)}{\prod\limits_{i=m+1}^{q}\Gamma^{\beta_i}(1-b_i+B_i s)\prod\limits_{j=n+1}^{p}\Gamma(a_j-A_j s)}\,z^s ds. \tag{6}$$

In (6), to compare with the definition of the I-function, part of the powers of the Γ-functions are taken to be equal to 1, as $\beta_i=1,\ i=1,...,m$ and $\alpha_j=1, j=n+1,...,p$.

When $\forall\,\alpha_j=\beta_i=1$, we have the better known Fox H-function (details can be seen, e.g., in [4]). But for non-integer values of these "power" parameters, the I-functions (4) and the $\overline{H}$-functions (6) do *not* reduce to Fox H-functions. Their scheme goes further, beyond the family of the "classical" special functions and of the special functions "of Fractional Calculus" as these have been assumed till recently.

According to [12] (Def. 6), the generalized Fox–Wright function (3) is representable in terms of the following I- and $\overline{H}$-functions:

$${}_p\widetilde{\Psi}_q\left[\begin{array}{c}(a_j,A_j,\alpha_j)_{j=1}^p\\(b_i,B_i,\beta_i)_{i=1}^q\end{array}\right|z\Big]=\overline{H}_{p,q+1}^{1,p}\left[-z\left|\begin{array}{c}(1-a_j,A_j,\alpha_j)_1^p\\(0,1),(1-b_i,B_i,\beta_i)_1^q\end{array}\right.\right]$$

$$=I_{p,q+1}^{1,p}\left[-z\left|\begin{array}{c}(1-a_j,A_j,\alpha_j)_1^p\\(0,1,1),(1-b_i,B_i,\beta_i)_1^q\end{array}\right.\right]. \tag{7}$$

Then, it is observed as a natural parallel with the representation of the Fox–Wright function (1) by a H-function:

$${}_p\Psi_q\left[\begin{array}{c}(a_1,A_1),...,(a_p,A_p)\\(b_1,B_1),...,(b_q,B_q)\end{array}\right|z\Big]={}_p\widetilde{\Psi}_q\left[\begin{array}{c}(a_1,A_1,1),...,(a_p,A_p,1)\\(b_1,B_1,1),...,(b_q,B_q,1)\end{array}\right|z\Big]$$

$$= H_{p,q+1}^{1,p}\left[-z\left|\begin{array}{c}(1-a_j,A_j)_1^p\\(0,1),(1-b_i,B_i)_1^q\end{array}\right.\right]. \tag{8}$$

We need to mention that in the initial work by Inayat-Hussain [14], as a particular case of the $\overline{H}$-function, another extension of the Fox–Wright function ${}_p\Psi_q$ has been considered, see Equation (27) in their work,

$${}_p\overline{\Psi}_q\left[\begin{array}{c}(a_j,1,\alpha_j)_1^p\\(b_i,1,\beta_i)_1^q\end{array}\right|z\left.\right] = H_{p,q+1}^{1,p}\left[-z\left|\begin{array}{c}(1-a_j,1,\alpha_j)_1^p\\(0,1),(1-b_i,1,\beta_i)_1^q\end{array}\right.\right]$$

However, our function ${}_p\widetilde{\Psi}_q$ is much more general than the one above, since the second components of the upper and low row parameters are taken only as $A_j = 1, B_i = 1$.

This paper is organized as follows. The definitions and a short discussion on the parameters of the considered special functions are given in Section 1. Section 2 is devoted to the basic properties of the introduced generalized Fox–Wright function (3). In Section 3, we evaluate the Laplace transform of the considered new special function, and survey some particular results. Section 4 provides the images of the generalized Fox–Wright function under the Erdélyi–Kober and Riemann–Liouville integrals of fractional order. In Section 5, numerous specific cases of the generalized Fox–Wright function (3) are considered, and we also emphasize their known applications in real-world models, thus justifying their origin and the needs of deep theoretical studies. Section 6 discusses in brief the role of the ${}_p\widetilde{\Psi}_q$-functions (in particular, of the multi-index Le Roy-type functions) as eigenfunctions of operators of new Fractional Calculus with Rathie I-functions as singular kernels, which we have introduced to extend our generalized fractional integrals involving Fox H-functions. Finally, in Section 7, we quickly comment on how our results work for the particular cases, quickly mention that some open problems are already stated in a previous survey [12], and discuss matters concerning numerical algorithms for classes of such special functions.

2. Basic Properties of the Generalized Fox–Wright Function

For the sake of simplicity and also for practical applications in real-world models, in what follows, we consider *all the parameters* a_j, A_j, α_j, b_i, B_i, β_i, *to be positive*. Then, the values $\Gamma(A_jk + a_j)$ or $\Gamma(B_ik + b_i)$ are also positive for all $k = 0, 1, 2, \ldots$ and we fix principal branches by taking $\Gamma^{\alpha_j}(A_jk + a_j) > 0$ and $\Gamma^{\beta_i}(B_ik + b_i) > 0$. Nevertheless, in previous works, some particular cases have been considered also when some of the parameters can be arbitrary real or even complex under suitable assumptions.

Theorem 1. *Let* ${}_p\widetilde{\Psi}_q$ *be the generalized Fox–Wright function, defined by the power series* (3) *with positive parameters* a_j, A_j, α_j, b_i, B_i, β_i, *and let* $\widetilde{\mu}$ *denote the basic characteristic parameter*

$$\widetilde{\mu} = 1 + \sum_{i=1}^{q}\beta_iB_i - \sum_{j=1}^{p}\alpha_jA_j. \tag{9}$$

Then, the following assertions hold true:

1. *If* $\widetilde{\mu} > 0$, *then the series* (3) *defines an entire function* (*that is, it is absolutely convergent for all* $z \in \mathbb{C}$).
2. *If* $\widetilde{\mu} = 0$, *then the series* (3) *defines an analytical function in the open disk* $D(0;\widetilde{R}) = \{z \in \mathbb{C} : |z| < \widetilde{R}\}$, *with the radius* $\widetilde{R} = \prod_{i=1}^{q} B_i^{\beta_iB_i} / \prod_{j=1}^{p} A_j^{\alpha_jA_j}$ (*that is, it is absolutely convergent for all* $z \in D(0;\widetilde{R})$).
3. *If* $\widetilde{\mu} < 0$, *then the series* (3) *converges only at the point 0.*

Proof. According to the Cauchy–Hadamard formula, the radius of convergence of the series (3) is $\widetilde{R} \geq 0$, where

$$\widetilde{R} = \frac{1}{\limsup\limits_{k\to\infty} |c_k|^{\frac{1}{k}}}, \quad \text{with} \quad c_k = \frac{\prod\limits_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod\limits_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{1}{k!}. \tag{10}$$

Applying Stirling's asymptotic formula for the Γ-function for large values of z ([8] Vol. 1) and after subsequent analytical manipulations (see, e.g., ([11] Rem. 6.5, (ii))), namely:

$$\Gamma(z+\alpha) \sim \sqrt{(2\pi)} z^{z+\alpha-\frac{1}{2}} \exp(-z), \quad |\arg(z+\alpha)| < \pi, \tag{11}$$

we obtain consecutively that

$$\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j) \sim \prod_{j=1}^{p} (\sqrt{2\pi})^{\alpha_j} \, (A_j k)^{\alpha_j (A_j k + a_j - 1/2)} \, e^{-\alpha_j A_j k},$$

$$\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i) \sim \prod_{i=1}^{q} (\sqrt{2\pi})^{\beta_i} \, (B_i k)^{\beta_i (B_i k + b_i - 1/2)} \, e^{-\beta_i B_i k},$$

$$\Gamma(k+1) \sim \sqrt{2\pi} \, z^{k+1/2} \, e^{-k}.$$

Then, due to the second formula in (10), it follows that

$$|c_k|^{1/k} \sim ABC_k \cdot k^{-\mu} \prod_{j=1}^{p} (A_j k)^{\alpha_j (a_j - 1/2)/k} \prod_{i=1}^{q} (B_i k)^{\beta_i (b_i - 1/2)/k} \, k^{-1/(2k)},$$

with

$$A = \prod_{j=1}^{p} (A_j^{\alpha_j A_j} \, e^{-\alpha_j A_j}); \quad B = e \prod_{i=1}^{q} (B_i^{-\beta_i B_i} \, e^{\beta_i B_i}); \tag{12}$$

and $C_k = (\sqrt{2\pi})^{-1/k} \prod\limits_{j=1}^{p} (\sqrt{2\pi})^{\alpha_j/k} \prod\limits_{i=1}^{p} (\sqrt{2\pi})^{-\beta_i/k}$. Thus,

$$|c_k|^{1/k} \sim AB \, k^{-\widetilde{\mu}}, \text{ when } k \to \infty, \tag{13}$$

i.e., $\widetilde{R} = \left(\lim\limits_{k\to\infty} |c_k|^{1/k} \right)^{-1} = (AB)^{-1} \lim\limits_{k\to\infty} k^{\widetilde{\mu}}$, and the rest immediately follows.

So, we can summarize that when $\widetilde{\mu} > 0$, $\widetilde{R} = \infty$, and ${}_p\widetilde{\Psi}_q$ is an entire function, while for $\widetilde{\mu} = 0$, it is an analytic one in an open disk centred at the origin with a finite radius $\widetilde{R}$, as given above. The series in (3) converges only at the origin, when $\widetilde{\mu} < 0$. □

It is useful to mention that if the parameters α_j and β_i are all equal to 1, then $\widetilde{\mu}$ and $\widetilde{R}$ in this theorem reduce to μ and R from (2). This is why, in this case, Theorem 1 is applicable to the classical Fox–Wright-generalized hypergeometric function (1).

In this article, we are mostly interested in the case in which the function (3) is entire, i.e., when $\widetilde{\mu}$ is positive. Before proceeding to further expositions, let us recall that an important characteristic of a given entire function f is the maximum of its modulus $M(r) = \max\limits_{|z|=r} |f(z)|$. More precisely ([16] Ch. 7, §1), if there exists a positive number ϑ such that $M(r) < \exp(r^{\vartheta})$, for all r sufficiently large, then f is said to be a function of a finite order $\rho = \inf \vartheta \geq 0$. Further, if f has a finite order ρ and there exists a positive number κ such that $M(r) < \exp(\kappa r^{\rho})$, then f is said to be a function of a finite type. The infimum of this κ for which the above inequality is valid for r sufficiently large is denoted by σ and

is called type f, namely, $\sigma = \inf \kappa \geq 0$. In view of the recalled definitions, the following asymptotic inequality holds true

$$| M(r) | < \exp((\sigma + \varepsilon) r^{\rho}), \quad \forall\, r > r_0(\varepsilon) > 0, \tag{14}$$

for each $\varepsilon > 0$ and r_0 sufficiently large.

The order and type of the entire function (3) (in the case $\widetilde{\mu} > 0$) are given by the following theorem.

Theorem 2. *Let ${}_p\widetilde{\Psi}_q$ be the generalized Fox–Wright function, defined by the formula* (3) *with positive parameters a_j, A_j, α_j, b_i, B_i, β_i, and let $\widetilde{\mu}$, defined by* (9)*, be positive. Then, the order $\widetilde{\rho}$ and type $\widetilde{\sigma}$ of the entire function* (3) *are represented by the following relations:*

$$\frac{1}{\widetilde{\rho}} = 1 + \sum_{i=1}^{q} \beta_i B_i - \sum_{j=1}^{p} \alpha_j A_j = \widetilde{\mu}, \tag{15}$$

respectively,

$$\frac{1}{\widetilde{\sigma}} = \widetilde{\rho} \cdot \Big(\prod_{j=1}^{p} A_j^{-\alpha_j A_j \widetilde{\rho}}\Big) \cdot \Big(\prod_{i=1}^{q} B_i^{\beta_i B_i \widetilde{\rho}}\Big) = \widetilde{\rho} \cdot \frac{\prod\limits_{i=1}^{q} B_i^{\beta_i B_i \widetilde{\rho}}}{\prod\limits_{j=1}^{p} A_j^{\alpha_j A_j \widetilde{\rho}}}, \tag{16}$$

i.e.,

$$\widetilde{\sigma} = \widetilde{\mu}\, \frac{\left(\prod\limits_{j=1}^{p} A_j^{\alpha_j A_j}\right)^{1/\widetilde{\mu}}}{\left(\prod\limits_{i=1}^{q} B_i^{\beta_i B_i}\right)^{1/\widetilde{\mu}}}. \tag{17}$$

Proof. The well-known formula expressing the order $\widetilde{\rho}$ of an entire function $\sum_{k=0}^{\infty} c_k z^k$ can be used, namely:

$$\widetilde{\rho} = \limsup_{k \to \infty} \frac{k \ln k}{\ln(1/|c_k|)}, \tag{18}$$

for calculating the order of (3). Applying Stirling's formula in the logarithmic form:

$$\ln \Gamma(z) = \left(z - \frac{1}{2}\right) \ln z + \frac{1}{2} \ln(2\pi) + O\left(\frac{1}{z}\right), \tag{19}$$

as well as the equality $\ln \dfrac{1}{|c_k|} = -\ln |c_k| = -\ln c_k$, the denominator in (18) is obtained. Indeed, beginning with

$$\ln c_k = \sum_{j=1}^{p} \ln[\Gamma^{\alpha_j}(A_j k + a_j)] - \sum_{i=1}^{q} \ln[\Gamma^{\beta_i}(B_i k + b_i)] - \ln[\Gamma(k+1)],$$

we have, respectively:

$$\ln[\Gamma(A_j k + a_j)] = \left(A_j k + a_j - \frac{1}{2}\right) \ln(A_j k + a_j) + \frac{1}{2} \ln(2\pi) + O\left(\frac{1}{k}\right),$$

$$\ln[\Gamma(B_i k + b_i)] = \left(B_i k + b_i - \frac{1}{2}\right) \ln(B_i k + b_i) + \frac{1}{2} \ln(2\pi) + O\left(\frac{1}{k}\right),$$

$$\ln \Gamma(k+1) = \left(k + \frac{1}{2}\right) \ln(k+1) + \frac{1}{2} \ln(2\pi) + O\left(\frac{1}{k}\right).$$

From the above formulae, it follows that

$$\ln c_k = \left[\sum_{j=1}^{p} \alpha_j A_j - \sum_{i=1}^{q} \beta_i B_i - 1\right] k \ln k + o(k \ln k),$$

whence

$$\frac{1}{\widetilde{\rho}} = \lim_{k\to\infty} \frac{\ln(1/|c_k|)}{k \ln k} = -\lim_{k\to\infty} \frac{\ln(c_k)}{k \ln k} = 1 + \sum_{i=1}^{q} \beta_i B_i - \sum_{j=1}^{p} \alpha_j A_j = \widetilde{\mu}.$$

Further, the type of the entire function $\sum_{k=0}^{\infty} c_k z^k$ of order $\widetilde{\rho}$ is expressed by the formula

$$(\widetilde{\sigma} e \widetilde{\rho})^{1/\widetilde{\rho}} = \limsup_{k\to\infty} \left(k^{1/\widetilde{\rho}} |c_k|^{1/k}\right). \tag{20}$$

Because of this, bearing in mind Formula (13), we obtain the right-hand side of (20), namely

$$\lim_{k\to\infty} \left(k^{1/\rho} |c_k|^{1/k}\right) = \lim_{k\to\infty} \left(k^{\widetilde{\mu}} AB\, k^{-\widetilde{\mu}}\right) = AB.$$

Now, (20) consequently gives

$$(\widetilde{\sigma} e \widetilde{\rho})^{1/\widetilde{\rho}} = AB = e^{\widetilde{\mu}} \cdot \Big(\prod_{j=1}^{p} A_j^{\alpha_j A_j}\Big) \cdot \Big(\prod_{i=1}^{q} B_i^{-\beta_i B_i}\Big),$$

$$(\widetilde{\sigma} \widetilde{\rho})^{1/\widetilde{\rho}} = \Big(\prod_{j=1}^{p} A_j^{\alpha_j A_j}\Big) \cdot \Big(\prod_{i=1}^{q} B_i^{-\beta_i B_i}\Big),$$

$$\widetilde{\sigma} = \frac{1}{\widetilde{\rho}} \cdot \Big(\prod_{j=1}^{p} A_j^{\alpha_j A_j \widetilde{\rho}}\Big) \cdot \Big(\prod_{i=1}^{q} B_i^{-\beta_i B_i \widetilde{\rho}}\Big),$$

which is equivalent to

$$\frac{1}{\widetilde{\sigma}} = \widetilde{\rho} \cdot \Big(\prod_{j=1}^{p} A_j^{-\alpha_j A_j \widetilde{\rho}}\Big) \cdot \Big(\prod_{i=1}^{q} B_i^{\beta_i B_i \widetilde{\rho}}\Big).$$

From the last formula, the equalities (16) and (17) automatically follow, which ends the proof of the theorem. □

By the general theory of entire functions (e.g., [16]), in particular according to formula (14), an upper asymptotic estimate is valid for the entire function (3). Namely, the following corollary can be formulated.

Corollary 1. *Let ${}_p\widetilde{\Psi}_q$ be the generalized Fox–Wright function, defined by the formula (3) with positive parameters a_j, A_j, α_j, b_i, B_i, β_i, and let $\widetilde{\mu}$, defined by (9), be positive. Then, there exists a positive number $r_0(\varepsilon) > 0$, depending only on ε, such that the following asymptotic estimation*

$$\left|{}_p\widetilde{\Psi}_q \left[\begin{matrix} (a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{matrix} \middle| z \right]\right| < \exp\left((\widetilde{\sigma} + \varepsilon)|z|^{\widetilde{\rho}}\right), \ \forall\, |z| > r_0(\varepsilon) > 0, \tag{21}$$

holds true, with $\widetilde{\rho}$ and $\widetilde{\sigma}$ as in (15) and (16).

3. Laplace Transform

We consider the Laplace transform of a given function f:

$$(\mathcal{L}f)(s) = \int_0^{\infty} f(t) e^{-st} dt, \ s \in \mathbb{C}.$$

If this integral is convergent at a point $s_0 \in \mathbb{C}$, then it converges absolutely for $s \in \mathbb{C}$ such that $\operatorname{Re}(s) > \operatorname{Re}(s_0)$. The number $\sigma_f = \inf \operatorname{Re}(s)$ of values s for which the Laplace integral converges is called the abscissa of convergence.

Now, we present results on the Laplace transform of the generalized Fox–Wright function (3). For the sake of convenience, we briefly denote it by ${}_p\widetilde{\Psi}_q(z)$, i.e.,

$$ {}_p\widetilde{\Psi}_q(z) := {}_p\widetilde{\Psi}_q\left[\begin{array}{c} (a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{array} \middle| z \right]. \tag{22}$$

Theorem 3. *Consider the generalized Fox–Wright function defined as in* (3) *and* (22)*, and let* $\widetilde{\mu}$ *be defined by* (9)*. Let also the parameters* a_j*,* A_j*,* α_j*,* b_i*,* B_i*,* β_i*,* α_0*,* β_0 *be positive, and the difference* $\widetilde{\mu} - \alpha_0 > 0$*, i.e.,*

$$1 + \sum_{i=1}^{q} \beta_i B_i - \sum_{j=1}^{p} \alpha_j A_j - \alpha_0 > 0. \tag{23}$$

Then, the Laplace transform $\mathcal{L}$ *of the product of the power function* $t^{\beta_0 - 1}$ *and the generalized Fox–Wright function* ${}_p\widetilde{\Psi}_q$ *is given by the formula*

$$\mathcal{L}\left\{ t^{\beta_0 - 1}\, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-\beta_0}\, {}_{p+1}\widetilde{\Psi}_q\left[\begin{array}{c} (a_j, A_j; \alpha_j)_{j=1}^{p+1} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{array} \middle| (\lambda s^{-\alpha_0}) \right], \tag{24}$$

with $a_{p+1} = \beta_0$*,* $A_{p+1} = \alpha_0$*, and* $\alpha_{p+1} = 1$*. In particular, if* $\beta_0 = 1$*, the formula* (24) *gives the Laplace transform image of the function* ${}_p\widetilde{\Psi}_q$ *only (without an additional multiplier). Namely,*

$$\mathcal{L}\left\{ {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-1}\, {}_{p+1}\widetilde{\Psi}_q\left[\begin{array}{c} (a_j, A_j; \alpha_j)_{j=1}^{p+1} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{array} \middle| (\lambda s^{-\alpha_0}) \right], \tag{25}$$

with $a_{p+1} = 1$*,* $A_{p+1} = \alpha_0$*, and* $\alpha_{p+1} = 1$*. Moreover, on the right half-plane* $\operatorname{Re}(s) > 0$*, the functions* (24) *and* (25) *are analytic.*

Proof. The function ${}_p\widetilde{\Psi}_q$ is an entire function of its argument for the assumed values of the parameters. Thus, interchanging the integral and the infinite sum in the expression below is justified. Bearing in mind that ${}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0})$ has the form

$$ {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) = \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{\lambda^k t^{\alpha_0 k}}{k!}, \tag{26}$$

we proceed as follows, calculating the left-hand side of (24). Namely, in view of (26), the chain of equalities below results as:

$$\mathcal{L}\left\{ t^{\beta_0 - 1}\, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = \int_0^{\infty} e^{-st}\, t^{\beta_0 - 1}\, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0})\, dt$$

$$= \int_0^{\infty} e^{-st}\, t^{\beta_0 - 1} \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{\lambda^k t^{\alpha_0 k}}{k!}\, dt$$

$$= \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{\lambda^k}{k!} \int_0^{\infty} e^{-st}\, t^{\alpha_0 k + \beta_0 - 1}\, dt$$

$$= \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{\lambda^k \, s^{-\alpha_0 k - \beta_0}}{k!} \int_0^{\infty} e^{-t} \, t^{\alpha_0 k + \beta_0 - 1} \, dt.$$

The above implies that the Laplace transform $\mathcal{L}\left\{ t^{\beta_0 - 1} \, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s)$ is equal to

$$s^{-\beta_0} \sum_{k=0}^{\infty} \frac{\Gamma(\alpha_0 k + \beta_0) \cdot \prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{(\lambda s^{-\alpha_0})^k}{k!}, \tag{27}$$

which is exactly the right-hand side of (24). Therefore,

$$\mathcal{L}\left\{ t^{\beta_0 - 1} \, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-\beta_0} \, {}_{p+1}\widetilde{\Psi}_q \left[\begin{matrix} (a_j, A_j; \alpha_j)_{j=1}^{p+1} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{matrix} \middle| (\lambda s^{-\alpha_0}) \right],$$

with $\alpha_{p+1} = 1$, $A_{p+1} = \alpha_0$, $a_{p+1} = \beta_0$. Finally, due to condition (23), the functions (24) and (25) are analytic functions in the whole complex plane, except for the point zero only. Therefore, in the right half-plane $\mathrm{Re}\,(s) > 0$, they both are analytic functions. □

Remark 1. *It is worth noting that, in general, the Laplace transform image of the generalized Fox–Wright function ${}_p\widetilde{\Psi}_q$ is the same kind of function but with the first subindex p increased to $p+1$, i.e., of the kind ${}_{p+1}\widetilde{\Psi}_q$, where the additional parameters are $A_{p+1} = \alpha_0$, $a_{p+1} = \beta_0$, and $\alpha_{p+1} = 1$.*

The cases in which α_0, β_0 coincide with a pair of the parameters, say $\alpha_0 = A_{j_0}$, $\beta_0 = a_{j_0}$ $(0 \le j_0 \le p)$, or alternatively, $\alpha_0 = B_{i_0}$, $\beta_0 = b_{i_0}$ $(0 \le i_0 \le q)$, are interesting. Then, the next two corollaries are valid.

Corollary 2. *Under the conditions of Theorem 3, and if additionally $\alpha_0 = A_{j_0}$, $\beta_0 = a_{j_0}$ for $0 \le j_0 \le p$, the following relation holds true:*

$$\mathcal{L}\left\{ t^{\beta_0 - 1} \, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-\beta_0} \, {}_p\widetilde{\Psi}_q \left[\begin{matrix} (a_j, A_j; \widetilde{\alpha}_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{matrix} \middle| (\lambda s^{-\alpha_0}) \right], \tag{28}$$

with $\widetilde{\alpha}_{j_0} = \alpha_{j_0} + 1$ and $\widetilde{\alpha}_j = \alpha_j$ when $j \ne j_0$. In particular, if $\beta_0 = 1$, the formula (28) takes the form

$$\mathcal{L}\left\{ {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-1} \, {}_p\widetilde{\Psi}_q \left[\begin{matrix} (a_j, A_j; \widetilde{\alpha}_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q} \end{matrix} \middle| (\lambda s^{-\alpha_0}) \right], \tag{29}$$

i.e., it gives the Laplace transform of the function ${}_p\widetilde{\Psi}_q$ only. Moreover, in the right half-plane $\mathrm{Re}\,(s) > 0$, the functions (28) and (29) are analytic.

Proof. Result (28) follows by setting $\alpha_0 = A_{j_0}$ and $\beta_0 = a_{j_0}$ in expression (27) and taking into account definition (3) of the generalized Fox–Wright function. Additionally, putting $\beta_0 = 1$ in the obtained formula, the relation (29) is immediately produced. The analyticities of the functions (28) and (29) follow as particular cases of Theorem 3. □

Corollary 3. *Under the conditions of Theorem 3, and if additionally $\alpha_0 = B_{i_0}$, $\beta_0 = b_{i_0}$, for $0 \le i_0 \le q$, the following relation holds:*

$$\mathcal{L}\left\{ t^{\beta_0 - 1} \, {}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0}) \right\}(s) = s^{-\beta_0} \, {}_p\widetilde{\Psi}_q \left[\begin{matrix} (a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \widetilde{\beta}_i)_{i=1}^{q} \end{matrix} \middle| (\lambda s^{-\alpha_0}) \right], \tag{30}$$

with $\widetilde{\beta}_{i_0} = \beta_{i_0} - 1$ *and* $\widetilde{\beta}_i = \beta_i$ *when* $i \neq i_0$. *In particular, if* $\beta_0 = 1$, *formula* (30) *transforms to the simpler form:*

$$\mathcal{L}\left\{{}_p\widetilde{\Psi}_q(\lambda t^{\alpha_0})\right\}(s) = s^{-1}\,{}_p\widetilde{\Psi}_{q-1}\left[\begin{array}{c|c} (a_j, A_j; \alpha_j)_{j=1}^{p} & \\ (b_i, B_i; \widetilde{\beta}_i)_{i=1}^{q-1} & (\lambda s^{-\alpha_0}) \end{array}\right], \tag{31}$$

with $(\widetilde{\beta}_1, \ldots, \widetilde{\beta}_{q-1}) = (\beta_1, \ldots, \beta_{i_0-1}, \beta_{i_0+1}, \ldots, \beta_q)$, *i.e., it gives the Laplace transform of the function* ${}_p\widetilde{\Psi}_q$. *Moreover, in the right half-plane* $Re\,(s) > 0$, *the functions* (30) *and* (31) *are analytic.*

Proof. The result (30) follows by setting $\alpha_0 = B_{i_0}$ and $\beta_0 = b_{i_0}$ in expression (27) and in view of definition (3) of the generalized Fox–Wright function. Further, taking into account that $\beta_0 - 1 = 0$ when $\beta_0 = 1$, formula (31) automatically follows. The analyticities of the functions (30) and (31) follow as particular cases of Theorem 3. □

4. Fractional-Order Integrals of the Generalized Fox–Wright Function ${}_p\widetilde{\Psi}_q$

Here, we briefly present some results on the images of the generalized Fox–Wright function (3) under the operators of Fractional Calculus (FC). A basic role in FC is played by the Riemann–Liouville (R-L) integral of fractional order $\delta > 0$ (e.g., [15]):

$$I^{\delta} f(z) = \frac{1}{\Gamma(\delta)} \int_0^z (z-\tau)^{\delta-1} f(\tau) d\tau = \frac{z^{\delta}}{\Gamma(\delta)} \int_0^1 (1-t)^{\delta-1} f(zt) dt, \tag{32}$$

which is assumed to be the identity operator for $\delta = 0$. Among the "classical" modifications of (32), with a very wide range of applications due to the bigger freedom of choice of the additional parameters, are the *Erdélyi–Kober (E-K) fractional integrals* [5,17], see also [15]:

$$I_{\eta}^{\nu,\delta} f(z) = z^{-\eta(\nu+\delta)} \int_0^z \frac{(z^{\eta} - \tau^{\eta})^{\delta-1}}{\Gamma(\delta)} \tau^{\eta\nu} f(\tau)\, d(\tau^{\eta})$$

$$= \frac{1}{\Gamma(\delta)} \int_0^1 (1-t)^{\delta-1} t^{\nu} f(zt^{\frac{1}{\eta}}) dt, \;\; \delta > 0\,(\text{order}),\; \nu \in \mathbb{R},\; \eta > 0, \tag{33}$$

where for $\delta = 0$ (by default), $I_{\eta}^{\nu,0} f(z) = f(z)$. Note that the E-K operator (33) preserves the power functions up to a constant multiplier, namely

$$I_{\eta}^{\nu,\delta}\{z^p\} = c_p\, z^p, \;\text{ with }\; c_p = \frac{\Gamma(p/\eta + \nu + 1)}{\Gamma(p/\eta + \nu + \delta + 1)}, \;\; p > -\eta(\nu+1), \tag{34}$$

and if we would like this to be valid for all $p = k = 0, 1, 2, \ldots$ (as for the members of a power series), the required condition becomes $\nu > -1$.

The *R-L integral* is included in (33) for the particular choice $\nu = 0, \eta = 1$, in the sense that

$$I^{\delta} f(z) = z^{\delta}\, I_1^{0,\delta} f(z).$$

Therefore, it is convenient to consider a more general form of the *"classical" operators of fractional integration* of E-K type, i.e.,

$$\mathbb{I} f(z) := z^{\delta_0}\, I_{\eta}^{\nu,\delta} f(z), \;\; \delta_0 \geq 0. \tag{35}$$

Theorem 4. *The image under the E-K fractional integral* $I_{\eta}^{\nu,\delta}$ *of the generalized Fox–Wright function* (3) *is again a function of the same kind but with subindices p and q increased by 1, that is, with* $p+1$ *and* $q+1$, *and with additional parameters arising from those of the E-K operator. Namely, for, real* $\lambda \neq 0$, $c > 0$, $\omega > 0$; $\delta > 0$, $\nu > -1$, $\eta > 0$; *and* $A_j > 0$, $B_i > 0$, $\alpha_j > 0$, $\beta_i > 0$, $j = 1, \ldots, p, i = 1, \ldots q$, *the following relation holds:*

$$I_{\eta}^{\nu,\delta}\left\{z^c\,{}_p\widetilde{\Psi}_q\left[\begin{array}{c|c} (a_j, A_j; \alpha_j)_{j=1}^{p} & \\ (b_i, B_i; \beta_i)_{i=1}^{q} & \lambda z^{\omega} \end{array}\right]\right\}$$

$$= z^c \, {}_{p+1}\widetilde{\Psi}_{q+1}\left[\begin{matrix}(a_j, A_j; \alpha_j)_{j=1}^{p}, \, (\nu+1+c/\eta, \omega/\eta, 1) \\ (b_i, B_i; \beta_i)_{i=1}^{q}, \, (\nu+\delta+1+c/\eta, \omega/\eta, 1)\end{matrix}\,\middle|\, \lambda z^\omega\right]. \tag{36}$$

And a similar result is obtained with a multiplier z^{δ_0} for an operator $\mathbb{I} f(z)$ of the form (35).

Proof. Since (3) is an entire function, we can exchange the order of the E-K integration and the summation for its power series. Thus, for $f(z) = z^c \, {}_p\widetilde{\Psi}_q\left[\begin{matrix}(a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q}\end{matrix}\,\middle|\, \lambda z^\omega\right]$, by term-by-term integration, we subsequently have:

$$I_\eta^{\nu,\delta} f(z) = \frac{1}{\Gamma(\delta)} \int_0^1 (1-t)^{\delta-1} \, t^\nu \left[z^c t^{\frac{c}{\eta}} \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \cdot \frac{\lambda^k z^{\omega k} \, t^{\frac{\omega k}{\eta}}}{k!} \right] dt$$

$$= z^c \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \, \frac{\lambda^k z^{\omega k}}{k!} \left[\frac{1}{\Gamma(\delta)} \int_0^1 (1-t)^{\delta-1} \, t^\nu \, t^{\frac{c}{\eta}} t^{\frac{\omega k}{\eta}} \, dt \right]$$

$$= z^c \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \Gamma^{\alpha_j}(A_j k + a_j)}{\prod_{i=1}^{q} \Gamma^{\beta_i}(B_i k + b_i)} \, \frac{\lambda^k z^{\omega k}}{k!} \left[\frac{1}{\Gamma(\delta)} \int_0^1 (1-t)^{\delta-1} \, t^{\nu + \frac{c}{\eta} + \frac{\omega k}{\eta}} \, dt \right].$$

Using the known formula for the Beta function, the above inner integral gives (for $\delta > 0$, $\nu > -1$, $\eta > 0$, $c > 0$, $\omega > 0$)

$$\frac{1}{\Gamma(\delta)} \cdot \frac{\Gamma(\delta) \, \Gamma(\nu + 1 + \frac{c}{\eta} + \frac{\omega k}{\eta})}{\Gamma(\nu + \delta + 1 + \frac{c}{\eta} + \frac{\omega k}{\eta})} = \frac{\Gamma^1(\frac{\omega}{\eta} k + \nu + 1 + \frac{c}{\eta})}{\Gamma^1(\frac{\omega}{\eta} k + \nu + \delta + 1 + \frac{c}{\eta})},$$

and so, the two additional $(p+1)$th and $(q+1)$th Γ-terms in the power series, and then

$$I_\eta^{\nu,\delta} \left\{ z^c \, {}_p\widetilde{\Psi}_q\left[\begin{matrix}(a_j, A_j; \alpha_j)_{j=1}^{p} \\ (b_i, B_i; \beta_i)_{i=1}^{q}\end{matrix}\,\middle|\, \lambda z^\omega\right] \right\}$$

$$= z^c \, {}_{p+1}\widetilde{\Psi}_{q+1}\left[\begin{matrix}(a_j, A_j; \alpha_j)_{j=1}^{p}, \, (\nu+1+c/\eta, \omega/\eta, 1) \\ (b_i, B_i; \beta_i)_{i=1}^{q}, \, (\nu+\delta+1+c/\eta, \omega/\eta, 1)\end{matrix}\,\middle|\, \lambda z^\omega\right].$$

Ending the proof, let us note that the condition $\widetilde{\mu} > 0$ as in (9) implies that the corresponding characteristic parameter $\widetilde{\mu^*}$ for the resulting ${}_{p+1}\widetilde{\Psi}_{q+1}$-function is also positive: $\widetilde{\mu^*} = \widetilde{\mu} + \frac{w}{\eta} - \frac{w}{\eta} = \widetilde{\mu} > 0$. That is, if the original ${}_p\widetilde{\Psi}_q$ is an entire function, the same holds for its E-K image ${}_{p+1}\widetilde{\Psi}_{q+1}$. In the case $\widetilde{\mu} = \widetilde{\mu^*} = 0$, the derived relation holds in the disc $|\lambda z^\omega| < \widetilde{R}$. □

In a way similar to that used by Kiryakova [18,19], etc., the results for the corresponding *E-K fractional derivatives* (see [5])

$$D_\eta^{\nu,\delta} f(z) = D_n \, I_\eta^{\nu+\delta, n-\delta} f(z)$$

$$= \prod_{j=1}^{n} \left(\frac{1}{\eta} z \frac{d}{dz} + \nu + j \right) I_\eta^{\nu+\delta, n-\delta} f(z), \quad n-1 < \delta \le n, \, n \in \mathbb{N}, \tag{37}$$

of ${}_p\widetilde{\Psi}_q$ can be written as well, and are presented by analogous expressions, again in terms of ${}_{p+1}\widetilde{\Psi}_{q+1}$.

Let us draw a parallel with the FC operator images in the case of the "classical" Fox–Wright function. If we put $\forall \alpha_j = 1, \beta_i = 1$, $j = 1, ..., p, i = 1, ..., q$, and also, for shortness, $c = 0$, the result (36) gives immediately this one from Kiryakova [18,19]:

$$I_\eta^{\nu,\delta}\left\{ {}_p\Psi_q\left[\begin{matrix} (a_1, A_1), \ldots, (a_p, A_p) \\ (b_1, B_1), \ldots, (b_q, B_q) \end{matrix} \middle| \lambda z^\omega \right] \right\}$$

$$= {}_{p+1}\Psi_{q+1}\left[\begin{matrix} (a_i, A_i)_1^p, (\nu + 1, \omega/\eta) \\ (b_j, B_j)_1^q, (\nu + \delta + 1, \omega/\eta) \end{matrix} \middle| \lambda z^\omega \right], \tag{38}$$

for $\delta > 0, \nu > -1, \eta > 0, \omega > 0, \lambda \neq 0$ and $\mu > 0$ as in (2). If $\mu = 0$, we require $|\lambda z^\omega| < R$, R as in (2). For the simpler particular case of the more popular ${}_pF_q$-function, the condition $\mu = 0 \Leftrightarrow p = q + 1$, and then $R = 1$.

Additionally, if $\nu = 0, \eta = 1$, this is the image under the R-L fractional integral, as from Kilbas ([20] Th. 2):

$$I^\delta\left\{ z^\nu {}_p\Psi_q\left[\begin{matrix} (a_j, A_j)_1^p \\ (b_i, B_i)_1^q \end{matrix} \middle| \lambda z^\omega \right] \right\}$$

$$= z^{\nu+\delta} {}_{p+1}\Psi_{q+1}\left[\begin{matrix} (a_j, A_j)_1^p, (\nu + 1, \omega) \\ (b_i, B_i)_1^q, (\nu + \delta + 1, \omega) \end{matrix} \middle| \lambda z^\omega \right], \ \delta > 0, \nu > -1.$$

To emphasize an interesting difference, we present below the E-K images of the two basic special functions defined by "fractional" parameters $\alpha > 0$ and $\gamma > 0$ but in a different way: for the *Mittag-Leffler function* $E_{\alpha,\beta}(z)$ (cf. [9,18]) and for the *Le Roy function* $F^\gamma(z)$ (see the next Section 5.2, (48)). Namely,

$$I_\eta^{\nu,\delta}\left\{E_{\alpha,\beta}(z)\right\} = I_\eta^{\nu,\delta}\left\{ \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)} \right\}$$

$$= I_\eta^{\nu,\delta}\left\{ {}_1\widetilde{\Psi}_1\left[\begin{matrix} (1,1,1) \\ (\beta, \alpha, 1) \end{matrix} \middle| z \right] \right\} = {}_2\widetilde{\Psi}_2\left[\begin{matrix} (1,1,1), (\nu + 1, 1/\eta, 1) \\ (\beta, \alpha, 1), (\nu + \delta + 1, 1/\eta, 1) \end{matrix} \middle| z \right],$$

while

$$I_\eta^{\nu,\delta}\{F^\gamma(z)\} = I_\eta^{\nu,\delta}\left\{ \sum_{k=0}^{\infty} \frac{z^k}{(k!)^\gamma} \right\}$$

$$= I_\eta^{\nu,\delta}\left\{ {}_1\widetilde{\Psi}_1\left[\begin{matrix} (1,1,1) \\ (1,1,\gamma) \end{matrix} \middle| z \right] \right\} = {}_2\widetilde{\Psi}_2\left[\begin{matrix} (1,1,1), (\nu + 1, 1/\eta, 1) \\ (1,1,\gamma), (\nu + \delta + 1, 1/\eta, 1) \end{matrix} \middle| z \right].$$

We need to note also that a result like (36) can be easily written *for compositions of E-K operators* with different parameters applied to ${}_p\widetilde{\Psi}_q$, say for operators of the form

$$\mathbb{I}_m f(z) = z^{\delta_0} I_{\eta_1}^{\nu_1,\delta_1} \ldots I_{\eta_m}^{\nu_m,\delta_m} f(z), \ \text{with different } \delta_i \geq 0, \nu_i, \eta_i, \ m = 1, 2, 3, \ldots, \tag{39}$$

and then the image of ${}_p\widetilde{\Psi}_q$ under such a *"generalized" FC operator* will be ${}_{p+m}\widetilde{\Psi}_{q+m}$.

For $m = 2$, a special case of (39) is the *Saigo fractional integral* with a Gauss hypergeometric function ${}_2F_1$ in the kernel, since it is representable also as a composition of *two* E-K integrals. For $m = 3$, such an operator is the *Marichev–Saigo–Maeda fractional integral*, whose kernel is the Appel F_3-function, and for arbitrary integer $m \geq 1$, we have the *hyper-Bessel integral operator* involving Meijer's $G_{m,m}^{m,0}$-function when written as a single integral operator of FC. Thus, by subsequent applications of result (36), it is easy to construct images of the generalized Fox–Wright function under all FC operators of this kind (39), as is achieved in [18,19] for the ${}_p\Psi_q$-functions in terms of the *generalized Fractional Calculus operators* (Kiryakova [5]).

5. Special Cases of the Generalized Fox–Wright Function ${}_p\widetilde{\Psi}_q$

By choosing different values of the parameters, many interesting and important special cases of the generalized Fox–Wright function ${}_p\widetilde{\Psi}_q$ can be obtained. Some of them are listed in this section. Depending on the parameters, two basic cases can be considered: where all the "power" parameters for the Gamma functions are positive integers, and where they are not necessarily so. Here, these are considered separately.

5.1. All the Parameters α_j, β_i are Positive Integers

Since some of the important special cases are defined only when all power parameters are positive integers, the case is separately considered. Then, the generalized Fox–Wright function ${}_p\widetilde{\Psi}_q(z)$, defined by (3), turns into a Fox–Wright generalized hypergeometric function of the kind of ${}_p\Psi_q$. It is worth noting that only then this happens. Indeed, if α_j and β_i are positive integers, say $\alpha_j = k_j \in \mathbb{N}$, and $\beta_i = l_i \in \mathbb{N}$, for $j = 1, \ldots, p$, $i = 1, \ldots, q$, then

$$ {}_p\widetilde{\Psi}_q\left[\begin{array}{c}(a_j, A_j; k_j)_{j=1}^{p}\\ (b_i, B_i; l_i)_{i=1}^{q}\end{array}\middle| z\right] = {}_K\Psi_L\left[\begin{array}{c}(\widetilde{a}_1, \widetilde{A}_1), \ldots, (\widetilde{a}_K, \widetilde{A}_K)\\ (\widetilde{b}_1, \widetilde{B}_1), \ldots, (\widetilde{b}_L, \widetilde{B}_L)\end{array}\middle| z\right], \tag{40} $$

with $K = k_1 + \cdots + k_p$, $L = l_1 + \cdots + l_q$, and

$$ (\widetilde{a}_1, \ldots, \widetilde{a}_K) = (\underbrace{a_1, \ldots, a_1}_{k_1 \text{ times}}, \ldots, \underbrace{a_p, \ldots, a_p}_{k_p \text{ times}}), $$

$$ (\widetilde{b}_1, \ldots, \widetilde{b}_L) = (\underbrace{b_1, \ldots, b_1}_{l_1 \text{ times}}, \ldots, \underbrace{b_q, \ldots, b_q}_{l_q \text{ times}}), $$

$$ (\widetilde{A}_1, \ldots, \widetilde{A}_K) = (\underbrace{A_1, \ldots, A_1}_{k_1 \text{ times}}, \ldots, \underbrace{A_p, \ldots, A_p}_{k_p \text{ times}}), $$

$$ (\widetilde{B}_1, \ldots, \widetilde{B}_L) = (\underbrace{B_1, \ldots, B_1}_{l_1 \text{ times}}, \ldots, \underbrace{B_q, \ldots, B_q}_{l_q \text{ times}}). $$

In particular, when $\alpha_j = 1$ and $\beta_i = 1$ ($j = 1, \ldots, p$, $i = 1, \ldots, q$), the function ${}_p\widetilde{\Psi}_q$ is the ${}_p\Psi_q$-function (1), i.e.,

$$ {}_p\widetilde{\Psi}_q\left[\begin{array}{c}(a_j, A_j; 1)_{j=1}^{p}\\ (b_i, B_i; 1)_{i=1}^{q}\end{array}\middle| z\right] = {}_p\Psi_q\left[\begin{array}{c}(a_1, A_1), ..., (a_p, A_p)\\ (b_1, B_1), ..., (b_q, B_q)\end{array}\middle| z\right]. \tag{41} $$

Typical elementary examples of (40) and (41) are the classical *Mittag-Leffler (M-L) functions* with one parameter (introduced by Mittag-Leffler himself, [21]), the two- and three-parameter M-L functions (the latter by Prabhakar [22]), and the Wright, Lorenzo–Hartley, Rabotnov, etc., functions. We do not feel it necessary to remind the unavoidable role and important applications of the Mittag-Leffler function, known as the Queen function of Fractional Calculus and of its simple extensions. We only recall below some of its definitions and presentations in terms of the Fox–Wright function:

$$ E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + 1)}, \; E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta b)}, \quad \alpha > 0, \; \beta > 0, $$

where

$$ E_{\alpha,\beta}(z) = {}_1\widetilde{\Psi}_1\left[\begin{array}{c}(1,1,1)\\ (\beta,\alpha,1)\end{array}\middle| z\right] = {}_1\Psi_1\left[\begin{array}{c}(1,1)\\ (\beta,\alpha)\end{array}\middle| z\right]. \tag{42} $$

The three-parametric variant, usually called the *Prabhakar function*, has an additional parameter τ, namely ([22]):

$$E^{\tau}_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{(\tau)_k}{\Gamma(\alpha k+\beta)} \frac{z^k}{k!} = \frac{1}{\Gamma(\tau)}\, {}_1\widetilde{\Psi}_1\left[\begin{array}{c} (\tau,1,1) \\ (\beta,\alpha,1) \end{array}\middle| z\right]$$
$$= \frac{1}{\Gamma(\tau)}\, {}_1\Psi_1\left[\begin{array}{c} (\tau,1) \\ (\beta,\alpha) \end{array}\middle| z\right], \quad \tau>0,\ \alpha>0,\ \beta>0, \tag{43}$$

where $(\tau)_k$ denotes the Pochhammer symbol: $(\tau)_k = \Gamma(\tau+k)/\Gamma(\tau)$ for $k \in \mathbb{N}$ and $(\tau)_0 = 1$.

Using in (40) and (41) particular values of the parameters and of the orders p and q, a long list of examples of known special functions can be obtained. Some of them are listed below.

Case 5.1.1. Let us take $m \in \mathbb{N}$, $p = m$, and $q = m+1$; then, for the following particular values of the parameters, we obtain that ${}_p\widetilde{\Psi}_q$ reduces to the *multi-index Mittag-Leffler function with* $3m$ *parameters* [23–26], namely:

$${}_m\widetilde{\Psi}_{m+1}\left[\begin{array}{c} (\tau_j,1;1)_{j=1}^{m} \\ (\beta_i,\alpha_i;1)_{i=1}^{m},(1,1,m-1) \end{array}\middle| z\right] = \Big[\prod_{i=1}^{m}\Gamma(\tau_i)\Big]\, E^{(\tau_i),m}_{(\alpha_i),(\beta_i)}(z). \tag{44}$$

And, if additionally all $\tau_j = 1$, it is the *multi-index Mittag-Leffler function with* $2m$ *parameters* [9,27–29]:

$${}_1\widetilde{\Psi}_m\left[\begin{array}{c} (1,1,1) \\ (\beta_i,\alpha_i,1)_{i=1}^{m} \end{array}\middle| z\right] = {}_1\Psi_m\left[\begin{array}{c} (1,1) \\ (\beta_i,\alpha_i)_{i=1}^{m} \end{array}\middle| z\right] = E^{m}_{(\alpha_i),\,(\beta_i)}(z) = E_{(\alpha_i),\,(\beta_i)}(z). \tag{45}$$

Both functions (44) and (45) can be considered as vector index generalizations of the Mittag-Leffler functions (43) and (42). These have been introduced and studied in our above-mentioned works by the powers series:

$$E^{(\tau_i),m}_{(\alpha_i),(\beta_i)}(z) = \sum_{k=0}^{\infty} \frac{(\tau_1)_k \dots (\tau_m)_k}{\Gamma(\alpha_1 k+\beta_1)\dots\Gamma(\alpha_m k+\beta_m)} \frac{z^k}{(k!)^m}, \tag{46}$$

and, respectively,

$$E_{(\alpha_i),\,(\beta_i)}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha_1 k+\beta_1)\dots\Gamma(\alpha_m k+\beta_m)}. \tag{47}$$

The multi-index generalization (47) of the M-L function (42) was introduced by Luchko et al. [30,31] and by Kiryakova [27] originally for positive parameters α_i, and studied extensively in our works [9,28,29]. Under weakened restrictions on the parameters α_i (or their real parts), not all necessarily positive, the study was extended by Kilbas, Koroleva and Rogosin [32]. Luchko (e.g., [30,31], etc.) also introduced and studied a multi-variable variant of the Mittag-Leffler-type functions which is widely explored in solving multi-term fractional-order differential equations. This is just one example of different multi-variable special functions related to operators and equations of fractional order, studied by numerous authors. Among the polynomials of many variables in this respect, we can also mention the 2D Appell polynomials, see [33]. The $3m$-index function (46) was introduced and studied by Paneva-Konovska [11,23] with larger parameter areas, generalizing both the Prabhakar function (43) and the ($2m$-)multi-index M-L functions (47). For more details on the parameter domains and properties and applications of these functions, see, e.g., the books [7,11] and also [9,10,26,29].

Case 5.1.2. If $m = 1$, then ${}_p\widetilde{\Psi}_q$ produces the M-L function (43) with three parameters, and if additionally $\tau_1 = 1$, it produces the M-L function (42) with two parameters.

Just to mention that the simplest particular cases of the ${}_p\Psi_q$-functions in Section 5.1, when all $A_j = \alpha_j = 1, B_i = \beta_i = 1,\ \ j = 1,\dots p, i = 1,\dots,q$, are the Gauss hypergeometric function ${}_2F_1$, the Kummer (confluent hypergeometric) function ${}_1F_1$ and the Bessel-type functions ${}_0F_1$, when $p = 2, 1, 0$ and $q = 1$.

More details and numerous particular cases of the multi-index Mittag-Leffler functions (46) and (47), and therefore of the Fox–Wright functions (40) and (41) and all included special functions of Fractional Calculus, are thoroughly described in our recent work "A Guide to Special Functions of Fractional Calculus" [10], see also [1,5,8,9,11,29]. These are Bessel-type functions, the Bessel–Clifford functions and their Wright two-, three- and four-index generalizations (Bessel–Wright and Lommel–Wright functions), as well as the hyper-Bessel functions (in the sense of Delerue). They are produced by letting $\tau_1 = \cdots = \tau_m = 1$ and $m \geq 2$, for a particular choice of the other parameters. In particular, the case $m = 2$ additionally provides some known classical special cases of (44) and (45), such as the Wright, Mainardi, Airy, Lommel, and Struve functions, all of these being cases of Dzrbashian's function [34] $\Phi_{1/\alpha_1,1/\alpha_2}(z;\beta_1,\beta_2) := E^{(2)}_{(\alpha_1,\alpha_2),(\beta_1,\beta_2)}(z)$, see details in [5,7,9–11], and so on.

5.2. The Parameters α_j, β_i Are Arbitrary Positive

At the beginning of the 20th century, the French mathematician Édouard Louis Emmanuel Julien Le Roy (1870–1954) introduced in [35] the following function (known as the *Le Roy function*):

$$F^{(\gamma)}(z) = \sum_{k=0}^{\infty} \frac{z^k}{[k!]^\gamma} = \sum_{k=0}^{\infty} \frac{z^k}{[\Gamma(k+1)]^\gamma}, \quad z \in \mathbb{C}, \tag{48}$$

with an arbitrary positive γ. This function appears to be a typical and even the simplest example of our generalized Fox–Wright function (3). Let us emphasize that in (48), the "fractional" index $\gamma > 0$ appears as the power of the Gamma function (factorial), while in the case of the Mittag-Leffler function (42), the "fractional" parameter $\alpha > 0$ is inside the argument of the Gamma function. The reasons for the author Le Roy to introduce (48) were quite similar to those of Mittag-Leffler himself for the function $E_\alpha(z)$, just for purely theoretical studies on analytical continuations of power series in the complex plane. But, it happens that (48) plays a practical role in describing various real-world processes. For example, a query has been raised by Kolokoltsov about the simplest case of the Le Roy function with index $\gamma = 1/2$: what kind of, if known, special function is this? Then, in paper [36], he recognized it (in Equation (51) in said paper) and emphasized that this function plays the same role for stochastic equations as the exponential and Mittag-Leffler functions for deterministic equations.

Recently, Gerhold [37] and, independently, Garra and Polito [38] have introduced the special function $F^{(\gamma)}_{\alpha,\beta}$ as an extension of both the M-L function $E_{\alpha,\beta}$ (42) and the Le Roy function (48), namely:

$$F^{(\gamma)}_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{[\Gamma(\alpha k+\beta)]^\gamma}, \quad z \in \mathbb{C}, \quad \text{with real positive } \alpha, \beta, \gamma. \tag{49}$$

The authors of [38] studied the above new function in relation to some integro-differential operators involving Hadamard fractional derivatives and hyper-Bessel-type operators. They emphasized the advantage of their approach by some examples, among which is a modified Lamb–Bateman equation. Later, this definition was extended by Garrappa, Rogosin and Mainardi in [39] under more general conditions for the parameters. There, they pointed out that the Mittag-Leffler–Le Roy function was involved in the solution of problems like the constructions of a Convey–Maxwell–Poisson distribution, which is important due to its ability to model count data with different degrees of over- and under-dispersion, and referred to Pogany [40]. Further studies on the function $F^{(\gamma)}_{\alpha,\beta}$ were performed also by Garra, Orsingher and Polito [41], then by Gorska, Horzela and Garrappa [42] and Simon [43], where these authors investigated the parameter conditions for its complete monotonicity. This special function is usually called *Le Roy-type function*, or the *Mittag-Leffler function of Le Roy type*, abbreviated to MLR.

A generalization of (49), involving one more index τ typical of the Prabhakar function (43), was introduced by Tomovski and Mehrez [44] while they studied families of generalized Mathieu-type power series and their associated probability distributions:

$$F^{(\gamma)}_{\alpha,\beta;\,\tau}(z) = \sum_{k=0}^{\infty} \frac{(\tau)_k}{[\Gamma(\alpha k+\beta)]^{\gamma}} \frac{z^k}{k!}, \quad z \in \mathbb{C},\ \tau \in \mathbb{C}, \tag{50}$$

where $(\tau)_k$ is the Pochhammer symbol ([8] Section 2.1.1): $(\tau)_0 = 1, (\tau)_k = \tau(\tau+1)\dots(\tau+k-1)$, if $k \in \mathbb{N}$. This may be called the *Le Roy function of Prabhakar type* (with four indices, abbreviated to MLPR) or the Prabhakar function of Le Roy type, and it is studied in detail by Paneva-Konovska [24,25] for extended conditions of the parameters.

Now, let us move on to the special cases of the function ${}_p\widetilde{\Psi}_q$, defined by (3) with subindices $p > 1$ and $q > 1$, taking the parameters α_j, β_i not necessarily to be integers. Under this condition, several examples are listed. Some rather general parameters are inspired by the multi-index Mittag-Leffler functions (46) and (47). One of them was considered first by Rogosin and Dubatovskaya [45,46], with $\alpha_i > 0$, $\beta_i > 0$, $\gamma_i > 0$, $i = 1, \dots, m$ (and also under more general conditions, e.g., with complex parameters satisfying $\sum_{i=1}^{m} \mathrm{Re}\,(\alpha_i\gamma_i) > 0$), called the *multi-parametric Le Roy function*. The next extension, which is a very general Le Roy-type special function, was introduced and studied in our recent works [47–49] also for complex parameters (including the condition $\sum_{i=1}^{m} \mathrm{Re}\,(\alpha_i\gamma_i) > 0$) as a *multi-index analogue of the previously mentioned functions*.

Case 5.2.1. If $m \in \mathbb{N}$, $p = m$, $q = m+1$, with a choice of the $4m$-parameters as below, ${}_p\widetilde{\Psi}_q$ produces the multi-index function, abbreviated to the *multi-MLPR function* $\mathbb{F}^{\gamma_i;m}_{\alpha_i,\beta_i;\tau_i}$ studied in our works [47–49],

$${}_m\widetilde{\Psi}_{m+1}\left[\begin{matrix}(\tau_j,1,1)_{j=1}^{m}\\ (\beta_i,\alpha_i,\gamma_i)_{i=1}^{m},(1,1,m-1)\end{matrix}\,\middle|\,z\right] = \Big[\prod_{i=1}^{m}\Gamma(\tau_i)\Big]\cdot \mathbb{F}^{\gamma_i;m}_{\alpha_i,\beta_i,\tau_i}(z). \tag{51}$$

If additionally all $\tau_j = 1$, then this ${}_m\widetilde{\Psi}_{m+1}$-function reduces to ${}_1\widetilde{\Psi}_m$, and this is the multi-parametric Le Roy function with $3m$ parameters, i.e.,

$${}_1\widetilde{\Psi}_m\left[\begin{matrix}(1,1,1)\\ (\beta_i,\alpha_i,\gamma_i)_{i=1}^{m}\end{matrix}\,\middle|\,z\right] = F^{(\gamma)_m}_{(\alpha,\beta)_m}(z), \tag{52}$$

as considered by Rogosin and Dubatovskaya [45]. The original definitions of these Le Roy-type functions by power series are recalled below:

$$\mathbb{F}^{\gamma_i;m}_{\alpha_i,\beta_i,\tau_i}(z) = \sum_{k=0}^{\infty} \frac{(\tau_1)_k\dots(\tau_m)_k}{\prod_{i=1}^{m}\Gamma^{\gamma_i}(\alpha_i k+\beta_i)} \cdot \frac{z^k}{(k!)^m}, \tag{53}$$

with $(\tau_j)_k$ being the Pochhammer symbol, and, respectively:

$$F^{(\gamma)_m}_{(\alpha,\beta)_m}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma^{\gamma_1}(\alpha_1 k+\beta_1)\dots\Gamma^{\gamma_m}(\alpha_m k+\beta_m)}. \tag{54}$$

We would like to emphasize that here, all the parameters γ_i are considered to be positive. However, regardless of this, we are free to consider γ_i to be real, and then we need the condition $\sum_{i=1}^{m} \alpha_i\gamma_i > 0$; i.e., the parameters of (52)–(54) satisfy the conditions

$$\tau_i > 0,\ \beta_i > 0,\ \alpha_i > 0,\ \gamma_i \in \mathbb{R}\ (i = 1,\dots,m),\ \text{ and } \sum_{i=1}^{m} \alpha_i\gamma_i > 0, \tag{55}$$

see [47–49] for more details.

Case 5.2.2. If $m = 1$, and with a special choice of parameters in Case 5.2.1, the ${}_p\widetilde{\Psi}_q$-function reduces to the MLPR function $F^{(\gamma)}_{\alpha,\beta,\tau}$ with four indices, given by (50), namely

$$ {}_1\widetilde{\Psi}_1\left[\begin{array}{c}(\tau,1,1)\\(\beta,\alpha;\gamma)\end{array}\middle|z\right] = \Gamma(\tau)\cdot F^{(\gamma)}_{\alpha,\beta,\tau}(z). \tag{56}$$

Additionally, if $\tau = 1$, (56) reduces to ${}_1\widetilde{\Psi}_1$, and this is the Le Roy function with three parameters, defined by (49), i.e.,

$$ {}_1\widetilde{\Psi}_1\left[\begin{array}{c}(1,1;1)\\(\beta,\alpha,\gamma)\end{array}\middle|z\right] = F^{(\gamma)}_{\alpha,\beta}(z). \tag{57}$$

Case 5.2.3. Consider now a case with $m = 2$. In paper [50], Pogany considered the problem for a closed-form definite integral expression for the COM–Poisson renormalization constant. There, the author mentioned, as an example only, a special function of the following form (here, we keep the original denotations but add one more parameter $\beta \neq \alpha$):

$$ F^{\alpha,\beta}_{(p,q;r,s)}(z) = \sum_{k=0}^{\infty} \frac{z^k}{[\Gamma(pk+q)]^{\alpha}[\Gamma(rk+s)]^{\beta}}\,. $$

This illustrates the case $m = 2$ of the multi-(2)-index Le Roy-type function with $4m = 8$ parameters, considered in our works [47–49]. It can be written in our denotations as

$$ \mathbb{F}^{(1,1);2}_{(p,r),(q,s);(\alpha,\beta)}(z) = \sum_{k=0}^{\infty} \frac{(1)_k(1)_k\, z^k}{(k!)^2\,[\Gamma(pk+q)]^{\alpha}[\Gamma(rk+s)]^{\beta}} = {}_1\widetilde{\Psi}_2\left[\begin{array}{c}(1,1,1)\\(q,p,\alpha),(s,r,\beta)\end{array}\middle|z\right]. $$

5.3. Cases of I- and $\overline{H}$-Functions Arising from Applications, Represented as the ${}_p\widetilde{\Psi}_q$-Function

As we mentioned in the Abstract, the generalized Fox–Wright ${}_p\widetilde{\Psi}_q$-function (3) is an important case of the I-functions of Rathie [13] and $\overline{H}$-functions of Inayat-Hussain [14] that are an extension to the Fox H-functions. Let us note that these functions, even if may they look unnecessarily general and complicated, appeared from the need to handle some real models, e.g., to include certain Feynman integrals from statistical physics, and some important mathematical functions—polylogarithms and the Riemann Zeta functions and its extensions.

Here, we give some details for the use of these I- and $\overline{H}$-special functions, and show that they all appear as typical and important cases of the ${}_p\widetilde{\Psi}_q$-function, which is the main object of our theoretical study.

Case 5.3.1. *Feynman integral*.

Rathie [13] introduced the I-function in the goals to cover some important functions of Applied Mathematics that are not included in the H-functions, among them are some Feynman integrals. Inayat-Hussain, in Part I of his paper [51], demonstrated the usefulness of such integrals in enabling the derivation of new transformation, summation and reduction formulae for single- and multiple-variable hypergeometric series. His results in Part II [14] evaluated certain Feynman integrals that arise in perturbation calculations of the equilibrium properties of a magnetic model of phase transitions in two ways. Namely, he studied the function

$$ g_1(z) = (-1)^m g(\gamma,\eta,\mu,m;z) = (2\pi)^{-d} \int\limits_{|p|\le 1} d\mathbf{p}|\mathbf{p}|^{2\eta-d}[\ln(1/|\mathbf{p}|)]^m|\mathbf{1}+z^{1/2}\mathbf{p}|^{-2\gamma}, $$

where z is real, g is the mentioned integral in Table 1, § 2.1 there, *m can even be a non-integer*, d is a dimension of the surface area of a unit sphere, and the constant $K_d = 2^{1-d}\pi^{-d/2}/\Gamma(d/2)$ is used below. In Equation (20) in his work, the author provides the following power series representation (with a constant multiplier $\widetilde{K} := K_{d-1}m!B(1/2,1/2+\mu/2)/2^{2+m}\pi$):

$$ g_1(z) = \widetilde{K}\sum_{k=0}^{\infty} \frac{(\gamma)_k(\gamma-\mu/2)_k\, z^k}{(1+\mu/2)_k(\eta+k)^{1+m}\,k!}\,. \tag{58}$$

We can rewrite this in the denotations of the $\widetilde{\Psi}$-function in the following way:

$$g_1(z) = \widetilde{K} \cdot \text{Const}_1 \sum_{k=0}^{\infty} \frac{\Gamma(\gamma+k)\,\Gamma(\gamma-\frac{\mu}{2}+k)\,[\Gamma(\eta+k)]^{1+m}}{\Gamma(1+\frac{\mu}{2}+k)\,[\Gamma(\eta+1+k)]^{1+m}}\,\frac{z^k}{k!}$$
$$= \text{Const}_2\ {}_3\widetilde{\Psi}_2\left[\begin{matrix}(\gamma,1,1),(\gamma-\mu/2),1,1),(\eta,1,1+m)\\(1_\mu/2,1,1),(\eta+1,1,1+m)\end{matrix}\,\middle|\,z\right]. \tag{59}$$

Let us recall that m can be a non-integer!

Case 5.3.2. *The Gaussian model of phase transitions in equilibrium statistical mechanics*.

The free energy of such a model on a Bravais lattice in d dimensions has been considered by Inayat-Hussain [14] and expressed in terms of the series (23) im that work, where the variable $\varepsilon = \beta_c/\beta - 1$ is a reduced temperature interval and $\beta_c = 2\xi/J$ is the critical temperature, as follows:

$$\beta F(d;\varepsilon) = -2^{-2-d}(1+\varepsilon)^{-2} \sum_{k=0}^{\infty} \frac{(1)_k\,[(3/2)_k]^d}{[(2)_k]^{1+d}\,(1+\varepsilon)^{2k}}.$$

In [14], Equation (28), the author presented this Gaussian model's free energy by the $\overline{H}$-function:

$$\beta F(d;\varepsilon) = -\frac{(1+\varepsilon)^{-2}}{4\pi^{d/2}}\,\overline{H}^{1,3}_{3,2}\left[-(1+\varepsilon)^{-2}\,\middle|\,\begin{matrix}(0,1,1),(0,1,1),(-1/2,1,d)\\(0,1,1)(-1,1,1+d)\end{matrix}\right].$$

Bearing in mind the above series representation, we can write this in terms of the $\widetilde{\Psi}$-function as follows:

$$\beta F(d;\varepsilon) = \text{const}\ {}_3\widetilde{\Psi}_2\left((1+\varepsilon)^{-2}\right) = \ldots = \text{const}\ {}_2\widetilde{\Psi}_1\left[\begin{matrix}(1,1,2),(3/2,1,d)\\(2,1,1+d)\end{matrix}\,\middle|\,(1+\varepsilon)^{-2}\right]. \tag{60}$$

Case 5.3.3. *The polylogarithm function.*

Consider the polylogarithm function with arbitrary index α (see, for example, ([8], Vol. 1), [52], etc.)

$$\text{Li}_\alpha(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^\alpha} = z + \frac{z^2}{2^\alpha} + \frac{z^3}{3^\alpha} + \cdots,\ \ |z|<1,\ \alpha \in \mathbb{C}, \tag{61}$$

which for particular choices of α appears as many other variants of the logarithm and polylogarithm functions. Note also that for (the singular value) $z = 1$ when $\alpha > 1$, this gives the *famous Riemann Zeta function*, $\text{Li}_\alpha(1) = \zeta(\alpha) = \sum\limits_{k=1}^{\infty} 1/(k^\alpha)$. The function (61) is also referred to as the *Jonquière function*. As in the other examples, one can mention some popular polynomials such as the Jonquière and Bernoulli polynomials and number sequences such as the Stirling and Eulerian numbers, etc.

In Section 7 of our survey [12], we have shown that this polylogarithm with fractional index α is representable not only as a Mittag-Leffler–Prabhakar function of Le Roy type, but more generally as the $\overline{H}$-function:

$$\text{Li}_\alpha(z) = -\overline{H}^{1,1}_{1,2}\left[-z\,\middle|\,\begin{matrix}(1,1,\alpha+1)\\(1,1,1),(0,1,\alpha)\end{matrix}\right].$$

Rewriting this as a power series gives the following $\widetilde{\Psi}$-function

$$\text{Li}_\alpha(z) = \sum_{k=1}^{\infty} \frac{\Gamma^\alpha(k)}{\Gamma^\alpha(k+1)}\,z^k = {}_2\widetilde{\Psi}_1\left[\begin{matrix}(0,1,\alpha),(1,1,1)\\(1,1,\alpha)\end{matrix}\,\middle|\,z\right]. \tag{62}$$

Case 5.3.4. *The generalized Riemann Zeta function (Hurwitz–Lerch Zeta function).*

One can consider also the Zeta-type function (see, e.g., [53]):

$$\Phi(z,\alpha,b) = \sum_{n=0}^{\infty} \frac{z^n}{(n+b)^\alpha} \tag{63}$$

$$= \overline{H}^{1,2}_{2,2}\left[-z\left|\begin{array}{c}(0,1,1),(1-b,1,\alpha)\\(0,1,1),(-b,1,\alpha)\end{array}\right.\right],\ b\neq 0,-1,-2,...,\ |z|\leq 1.$$

This contains not only the Riemann Zeta function $\zeta(\alpha)$ (for $z=1$, $b=0$) and the Hurwitz Zeta function $\zeta(\alpha,b)=\Phi(1,\alpha,b)$ with $\mathrm{Re}\,(\alpha)>1$, but also the above-mentioned polylogarithm function (61), as $\mathrm{Li}_\alpha(z)=z\,\Phi(z,\alpha,1)$, $\alpha\in\mathbb{C}$, when $|z|<1$, or $\mathrm{Re}\,(\alpha)>1$ when $|z|=1$. The more general Hurwitz–Lerch Zeta function is:

$$\Phi^{\rho,\sigma}_{\mu,\nu}(z,\alpha,b)=\sum_{n=0}^{\infty}\frac{(\mu)_{\rho n}}{(\nu)_{\sigma n}}\cdot\frac{z^n}{(n+b)^\alpha},\ \text{for}\ |z|<\delta=\rho^{-\rho}\sigma^{\sigma},$$

involving Pochhamer symbols. We omit here the details of the parameters and of the contours of Mellin–Barnes-type integrals for this function, as well as for the more general case below, referring to [53].

In [53], Srivastava, Saxena, Pogány and Saxena considered *the generalized Hurwitz–Lerch Zeta function*

$$\Phi^{(\rho,\sigma,\kappa)}_{\lambda,\nu,\mu}(z,\alpha,b)=\sum_{n=0}^{\infty}\frac{(\lambda)_{\rho n}(\mu)_{\sigma n}}{(\nu)_{\kappa n}\,n!}\cdot\frac{z^n}{(n+b)^\alpha}. \tag{64}$$

As we mentioned in [12], Section 7, this can be written in terms of an $\overline{H}$ function, but now we give its explicit form as a $\widetilde{\Psi}$-function. Namely,

$$\Phi^{(\rho,\sigma,\kappa)}_{\lambda,\nu,\mu}(z,\alpha,b)=\frac{\Gamma(\nu)}{\Gamma(\lambda)\Gamma(\mu)}\,\overline{H}^{1,3}_{3,3}\left[-z\left|\begin{array}{c}(1-\lambda,\rho,1),(1-\mu,\sigma,1),(1-b,1,\alpha)\\(0,1),(1-\nu,\kappa,1),(-b,1,\alpha)\end{array}\right.\right]$$

$$=\text{const}\ {}_3\widetilde{\Psi}_2\left[\begin{array}{c}(\lambda,\rho,1),(\mu,\sigma,1),(b,1,\alpha)\\(\nu,\kappa,1),(1+\beta,1,\alpha)\end{array}\right|z\Bigg]. \tag{65}$$

As a conclusion of this section, we would like to note that all the functions listed above can be now associated with the class of the so-called *special functions of Fractional Calculus* (SFs of FC), bearing in mind the arguments and results stated in the previous Section 4 and in the next Section 6.

6. The Role of the New Special Functions as Eigenfunctions of New Fractional Calculus Operators

When introducing and studying some new classes of special functions (SFs), one important point to address, so to justify their necessity, is determining the corresponding linear integral L/differential operators D that transform such functions f into themselves just by multiplying by a scalar, e.g., $Df(z)=\lambda f(z)$, $\lambda\neq 0$. For shortness, we call such operators *"eigen"-operators* of the function f. The reasons for this are to show that an introduced SF may appear to be a solution of a corresponding integral/differential equation with this eigen-operator. For the SFs of FC, these should be an integral/integro-differential operator *of fractional order (or multi-order)*, or at least some analogue of more general operators of FC.

For the case of typical SFs of FC, such as the Mittag-Leffler function (42), the answer is well known. For example, the Rabotnov (α-exponential) function $y_\alpha(z)=z^{\alpha-1}E_{\alpha,\alpha}(\lambda z)$ satisfies the Riemann–Liouville FODE $D^\alpha\,y_\alpha(z)=\lambda\,y_\alpha(z)$. More generally, for $E_{\alpha,\beta}(z)$, the eigen-operators are Erdélyi–Kober fractional integrals (33)/derivatives (37) ([5], Ch. 2). For the $2m$ multi-index Mittag-Leffler function defined by (47), the corresponding eigen-operators are represented in terms of the generalized FC operators in the sense of [5], the so-called multiple Erdélyi–Kober fractional integrals/derivatives, see [9,27,54].

A useful tool we have used to find out and to represent the eigen-operators for the above-mentioned class of Mittag-Leffler-type functions is the theory of *Gelfond–Leontiev (G-L) operators of generalized integration and differentiation* generated by (and with respect to) the corresponding special function. These operators are constructed by means of a power series involving the coefficients of the entire function, as the considered SFs are. Details on the definition of the G-L operators and their representation by means of operators of

FC can be found in our works on M-L-type functions [9,27,54] and recently for some Le Roy-type functions in [12].

Shortly, based of the definition (3) of the generalized Fox–Wright function

$$ {}_p\widetilde{\Psi}_q\left[\begin{array}{c}(a_j,A_j;\alpha_j)_{j=1}^{p}\\(b_i,B_i;\beta_i)_{i=1}^{q}\end{array}\middle|z\right]=\sum_{k=0}^{\infty}\frac{\prod_{j=1}^{p}\Gamma^{\alpha_j}(A_jk+a_j)}{\prod_{i=1}^{q}\Gamma^{\beta_i}(B_ik+b_i)}\cdot\frac{z^k}{k!}=\sum_{k=0}^{\infty}c_kz^k, $$

$$ \text{with}\quad c_k=\frac{\prod_{j=1}^{p}\Gamma^{\alpha_j}(A_jk+a_j)}{k!\prod_{i=1}^{q}\Gamma^{\beta_i}(B_ik+b_i)}, \tag{66} $$

the corresponding G-L operator of generalized integration with respect to ${}_p\widetilde{\Psi}_q$ can be defined for an analytic function $f(z)=\sum_{k=0}^{\infty}a_kz^k$ by the convergent power series

$$ L_{\widetilde{\Psi}}f(z)=\sum_{k=0}^{\infty}a_k\,z^{k+1}\,\frac{c_{k+1}}{c_k} $$

$$ =\sum_{k=0}^{\infty}a_k\,z^{k+1}\,\frac{\prod_{j=1}^{p}\Gamma^{\alpha_j}(A_jk+A_j+a_j)\cdot\prod_{i=1}^{q}\Gamma^{\beta_i}(B_ik+b_i)}{(k+1)\prod_{j=1}^{p}\Gamma^{\alpha_j}(A_jk+a_j)\cdot\prod_{i=1}^{q}\Gamma^{\beta_i}(B_ik+B_i+b_i)}. \tag{67} $$

It happens that, at least at this stage, the operator (67) can hardly be represented and interpreted as some integral operator in the most general case of ${}_p\widetilde{\Psi}_q$. However, as illustrated in our previous work [12], a construction of the eigen-operator for one of the important cases of the generalized Fox–Wright function (3) is performed successfully and it appears as a kind of new generalized integral operator of fractional multi-order.

Namely, this is achieved in the case of the *multi-parametric Le Roy function of Mittag-Leffler type* $F^{(\gamma)_m}_{(\alpha,\beta)_m}(z)$ in Section 5, Case 5.2.1, denoted here for convenience by $F^{(\beta)_m}_{(B,b)_m}(z)$. As shown in [12] (Equations (54)–(56)), this special function (52), (54) is representable as a generalized Fox–Wright function (3), and thus also as $\overline{H}$- and I-functions (4). Namely,

$$ F^{(\beta)_m}_{(B,b)_m}(z)=\sum_{k=0}^{\infty}\frac{z^k}{\prod_{i=1}^{m}\Gamma^{\beta_i}(B_ik+b_i)}={}_1\widetilde{\Psi}_m\left[\begin{array}{c}(1,1,1)\\(b_i,B_i,\beta_i)_1^m\end{array}\middle|z\right] \tag{68} $$

$$ =\overline{H}^{1,1}_{1,m+1}\left[-z\middle|\begin{array}{c}(0,1,1)\\(0,1,1),(1-b_i,B_i,\beta_i)_1^m\end{array}\right] $$

$$ =I^{1,1}_{1,m+1}\left[-z\middle|\begin{array}{c}(0,1,1)\\(0,1,1),(1-b_i,B_i,\beta_i)_1^m\end{array}\right],\quad m\geq 1. \tag{69} $$

For $F^{(\beta)_m}_{(B,b)_m}(z)$, the corresponding G-L operators of generalized differentiation and integration of an analytic function $f(z)=\sum_{k=0}^{\infty}a_kz^k$ have the simpler forms

$$ \mathbf{D}\,f(z):=D_{mMLR}\,f(z)=\sum_{k=1}^{\infty}a_kz^{k-1}\cdot\prod_{i=1}^{m}\frac{\Gamma^{\beta_i}(B_ik+b_i)}{\Gamma^{\beta_i}(B_ik+b_i-B_i)}, \tag{70} $$

respectively,

$$ \mathbf{L}\,f(z):=L_{mMLR}\,f(z)=\sum_{k=0}^{\infty}a_k\,z^{k+1}\cdot\prod_{i=1}^{m}\frac{\Gamma^{\beta_i}(B_ik+b_i)}{\Gamma^{\beta_i}(B_ik+b_i+B_i)}. \tag{71} $$

Evidently, $\mathbf{D}\,\mathbf{L}\,f(z)=f(z)$.

Theorem 5 from [12] is stated as follows.

Theorem 5. *The generalized G-L operator of differentiation* (70) *is an eigen-operator for the function* (52), (54)*; that is,* $F^{(\beta)_m}_{(B,b)_m}$ *is an eigenfunction for this (G-L) "differentiation" operator:*

$$\boldsymbol{D}\, F^{(\beta)_m}_{(B,b)_m}(\lambda z) = D_{mMLR}\, F^{(\beta)_m}_{(B,b)_m}(\lambda z) = \lambda\, F^{(\beta)_m}_{(B,b)_m}(\lambda z),\ \ \lambda \neq 0. \tag{72}$$

For G-L integration (71)*, the corresponding relation has the form:*

$$\boldsymbol{L}\, F^{(\beta)_m}_{(B,b)_m}(\lambda z) = L_{mMLR}\, F^{(\beta)_m}_{(B,b)_m}(\lambda z) = \frac{1}{\lambda}\, F^{(\beta)_m}_{(B,b)_m}(\lambda z) - \frac{1}{\lambda \prod\limits_{i=1}^{m} \Gamma^{\beta_i}(b_i)},\ \ \lambda \neq 0. \tag{73}$$

Both relations (72) and (73) are just close *analogues of these for the multi-index M-L functions* and their eigen-operators, as in [9,27,54].

For the G-L integral (71), we have found also an integral representation (Theorem 6, [12]) as an analogue of the generalized fractional integration operator with H-function $H^{m,0}_{m,m}$ as a kernel. But instead, we have now a Rathie I-function, and can consider it again as *an integral operator thought as an operator of "fractional multi-order"*!

Theorem 6. *The G-L integration operator* (71) *of an entire function* $f(z)$*, generated by means of the Le Roy-type function* (52), (54)*, can also be represented by means of the integral operator*

$$\mathbb{I}^m f(z) = L_{mMLR}\, f(z) = z \int_0^1 I^{m,0}_{m,m}\left[\sigma \middle| \begin{matrix} (b_i, B_i, \beta_i)_1^m \\ (b_i - B_i, B_i, \beta_i)_1^m \end{matrix}\right] f(z\sigma)d\sigma. \tag{74}$$

This can be interpreted as some generalized fractional integration of multi-order $(B_1 > 0, ..., B_m > 0)$*. It has also an equivalent form as a commutable composition of what we call "Le Roy-type" Erdélyi–Kober integrals* (*analogues of* (33)) *of the form*

$$\mathbb{I}^1_i f(z) = \int_0^1 I^{1,0}_{1,1}\left[\sigma \middle| \begin{matrix} (b_i, B_i, \beta_i) \\ (b_i - B_i, B_i, \beta_i) \end{matrix}\right] f(z\sigma)d\sigma,\ \ i = 1, ..., m. \tag{75}$$

Namely,

$$\mathbb{I}^m f(z) = z\left[\prod_{i=1}^{m} \mathbb{I}^1_i\right] f(z) = z\, \mathbb{I}^1_m \left\{ \mathbb{I}^1_{m-1} \cdots \left[\mathbb{I}^1_1 f(z)\right] \right\}. \tag{76}$$

We shall call the operators (71) and (74) the *Gelfond–Leontiev–Le Roy (G-L-Le Roy) operators of generalized integration*.

We note an analogy with the M-L ($m = 1$) and multi-index M-L functions ($m \geq 1$) and their corresponding eigen-operators, being *operators of Generalized Fractional Calculus* (multiple Erdélyi–Kober operators [5]). Namely, with $\forall \beta_i = 1$, for (47), we have the eigenfunction relation

$$D_{mML}\, E_{(B_i),(b_i)}(\lambda z) = \lambda\, E_{(B_i),(b_i)}(\lambda z),\ \ \lambda \neq 0,$$

where

$$D_{mML}\, f(z) = \frac{1}{z}\left[D^{(b_i-1),(B_i)}_{(1/B_i),m} - \text{const} \cdot f(0)\right],$$

and its right inverse is a *generalized fractional integral* of multi-order $(B_1, ..., B_m)$ of the form

$$L_{mML}\, f(z) = z\, I^{(b_i-1),(B_i)}_{(1/B_i),m}\, f(z) = z \int_0^1 H^{m,0}_{m,m}\left[\sigma \middle| \begin{matrix} (b_i, B_i)_1^m \\ (b_i - B_i, B_i)_1^m \end{matrix}\right] f(z\sigma)d\sigma. \tag{77}$$

See details in [9,10,27,54].

In a following work, we will introduce and study *new generalized fractional integrals* with $I^{m,0}_{m,m}$-kernels and more general parameters than in (74), for which the semigroup property and other basic axioms of Fractional Calculus are satisfied. These will be a further extension of the pioneering ideas of Kalla [55] to consider "generalized fractional integrals" whose kernels can be various special functions, like the Bessel and Gauss functions and Meijer's $G^{m,n}_{p,q}$- and Fox's $H^{m,n}_{p,q}$-functions. Our generalized Fractional Calculus in [5] was

based on the use of the $G^{m,0}_{m,m}$ and $H^{m,0}_{m,m}$-functions, so the next new step is to involve the Rathie I-functions.

Just to raise slightly the curtain, a more general form of the integral operator (75) for $\underline{m=1}$, as a further analogue of the Erdélyi–Kober integral, depending on four (instead on three) parameters, can be considered in the form:

$$\mathbb{I}^{\mu,\alpha}_{\beta,\gamma} f(z) = \int_0^1 I^{1,0}_{1,1}\left[\sigma \left| \begin{matrix} (\mu+\alpha-\beta,\beta,\gamma) \\ (\mu-\beta,\beta,\gamma) \end{matrix} \right.\right] f(z\sigma)d\sigma,$$

for order $\alpha > 0$, and for $\alpha = 0$: $\mathbb{I}^{\mu,0}_{\beta,\gamma} f(z) := f(z)$. One can check that the semi-group property is then satisfied, and it has the form:

$$\mathbb{I}^{\mu+\alpha_1,\alpha_2}_{\beta,\gamma}\, \mathbb{I}^{\mu,\alpha_1}_{\beta,\gamma} = \mathbb{I}^{\mu,\alpha_1+\alpha_2}_{\beta,\gamma}, \ \alpha_1 > 0, \alpha_2 > 0,$$

which is very similar to that of the "classical" E-K fractional integral. The case of arbitrary $m \geq 1$ is to be considered in our following work.

7. Conclusions

7.1. Corollaries for the Particular Cases

All the results in Sections 2, 3, 4 and 6 are applicable to the particular cases mentioned in Section 5. For example, the new Theorems 1, 2, 3 and 4 stated here, and their corollaries, are valid for the listed cases with the corresponding $\widetilde{\mu}$, $\widetilde{R}$, expressed by particular parameter values. Note that when the corresponding parameter $\widetilde{\mu} > 0$, each of them is an entire function. In particular, the results related to the Laplace transform and to the fractional-order integrals in Sections 3 and 4 are new, and are valid in particular for the Mittag-Leffler and Le Roy-type functions and their multi-index extensions, among all other classical and FC special functions. Theorems 5 and 6 reveal the relation between the considered special functions and the new generalized operators of Fractional Calculus.

We have to note that the results for the function ${}_p\Psi_q$ in (1) are well known or already available in our previous papers. Therefore, the results for the functions in Cases 5.1.1 and in Case 5.1.2, when the parameters α_j and β_i are positive integers, automatically follow just as corollaries. The functions in Cases 5.2.1 and 5.2.2, when α_j and β_i are arbitrary positive, have been recently introduced and also discussed for more general domains of the parameters (for details, see, e.g., the papers [12,24,25,45–49]). The examples in Cases 5.3 show the wide scope of the introduced Fox–Wright-type functions ${}_p\widetilde{\Psi}_q$, and justify the reasons for the theoretical studies on their analytical properties.

7.2. On Some Open Problems

We would like to refer the readers to Section 8 in our survey [12] for several open problems related to the studies of the Le Roy-type functions and of the Rathie I- and Inayat-Hussain $\overline{H}$-functions. The same problems concern the generalized Fox–Wright functions ${}_p\widetilde{\Psi}_q$ treated here, as they encompass the mentioned special functions.

7.3. Remarks on Numerical Algorithms for Some General Classes of Special Functions

Some doubts may arise regrading whether analytical studies on such general special functions with many parameters are useful nowadays, with the existence of computers and various CASs. In brief, let us refer to Section 9 of survey [10], where we have cited the opinions of some well-known promoters of special functions on this matter.

First, let us cite Richard Askey: "... The advent of fast computing machines was thought to have made special functions a subject of the past. The reality has been different. Continued development of older functions and *the introduction of new special functions has been the reality*. ... and still remains a lot to be discovered .... The classical handbooks ... although useful as references, maybe no longer enough as primary means of accessing the special functions of mathematical physics. A number of high level programs appeared that

are better suited for this challenging purpose (to handle more complicated SF), to mention Mathematica, Maple, Matlab, Mathcad, ..."

Then, we may recall the problems that Stephen Wolfram, the founder of the Wolfram project, shared about the situation they came across in beginning, with the pessimism of a US government lab. Then, he was told that: "... Look, you have to understand that by the end of the 1990s we hope to have the integer-order Bessel functions done to quad precision ... You know, it's actually quite a difficult thing to put a special function into Mathematica. You don't just have to do the numerics ... *So what makes a special function good?*" Then, he (S.W.) continued, " ... There gradually started to appear systematic reference works on the properties of special functions. Each one based on lot of work ...", "... I guess integrals (meant as the analytical studies ...) are timeless. They don't really bear the marks of the human creators. So we have the tables, but we really don't quite know where they came from ...".

Here, we would like to mention Marichev's book [52] containing the basic theory on which the Wolfram project was based. Also, we refer to a recent preprint by Marichev and Shishkina [56]. In both these sources, the main idea is first to represent a classical or new special function in terms of the general Meijer's G-function and then to apply a routine for G-functions. This arXiv survey sheds light on the numerical procedures by Wolfram Mathematica developed *for G- and H-functions and prospective, more general ones*. As the authors say there: "In recent years Wolfram Language has added some new generic functions, as Heun functions, Lame functions, Carlson elliptic integrals, *Fox H-function*, ...". Well, the next task can be to implement this in CAS functions, like those of Le Roy type, Riemann Zeta functions, cases of I- and $\overline{H}$-functions, etc., by using the power series representation of the generalized Fox–Wright function ${}_p\widetilde{\Psi}_q$ and its analytical properties.

Concerning some "simpler" special functions of Fractional Calculus, like the Mittag-Leffler, Prabhakar and some Wright functions (including multi-index Mittag-Leffler functions with 2×2 indices, i.e., $m = 2$), we would like to refer the readers to Section 9 of [10] for some brief information on numerical procedures by Maple, Matlab, and Mathematica, proposed by the following authors: Caputo and Mainardi; Gorenflo, Loutchko and Luchko; Diethem, Ford, Freed and Luchko; Podlubny; Hilfer and Seybold; Garrappa; Popolizio; Luchko, Trujillo and Velasco; Consiglio; Mainardi and Masina; etc. And, recently, some numerical experiments have been performed for the Mittag-Leffler–Le Roy function related to the conditions for complete monotonicity in paper [42] by Gorska, Horzela and Garrappa.

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